

An Unsupervised Search for Morphological Anomalies of Fast Radio Bursts From Tabulated Parameters of The Second CHIME/FRB Catalog

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Abstract

The second CHIME/FRB catalogue is the largest fast radio burst (FRB) sample released to date. Morphology is primarily used to characterise the population rather than to flag individually atypical bursts. Understanding structure of single bursts could provide insight into the origin or mechanisms of FRBs. This work ran an unsupervised search over eight tabulated shape parameters. Six non-parametric detectors were used, chosen to span a range of anomaly types. Sensitivity was quantified by injecting synthetic anomalies over a range of strengths; specificity testing rejected indiscriminate detectors. The result is a shortlist of 36 morphologically atypical candidates, annotated with the driving parameter and fit reliability. The majority of the shortlist is driven by parameters dependent on instrumental bias. The shortlist stands as a set of bursts whose dynamic spectra warrant direct examination.

CONTENTS

1	Introduction	1
2	Method	1
	2.1 Cleaning	2
	2.2 Detectors	2
	2.3 Injection Recovery	3
	2.4 Candidate Validation	4
	2.5 Anomaly Context	5
3	Results	5
4	Discussion	9
5	Conclusion	12
	Appendix A	d
	Appendix B	e

1 Introduction

Fast radio bursts (FRBs) are highly energetic, transient radio emissions (Petroff et al. 2019). Lasting milliseconds, the duration bounds the emitting region of these objects to a few hundred kilometres. Furthermore, FRBs are some of the most luminous emissions in the sky with an implied brightness temperature on the order of 10^{35}K for a one-jansky millisecond burst (Cordes & Chatterjee 2019). This is twelve orders of magnitude above that of pulsars at kiloparsec distances.

FRBs present as repeaters or non-repeaters, the latter forming the bulk of the catalogued population (Collaboration et al. 2026). but this results in a dichotomy of origin. A non-repeating FRB suggests a cataclysmic event within a stellar body. However, a repeating source implies the progenitor survives the large energy release. This asymmetry has led morphological research to probe whether repeating and non-repeating FRBs are the same objects (Pleunis et al. 2021). But this framing does not explore if the morphological shape of the burst, is itself, atypical which could allow for further insight into the mechanism and progenitor.

Independent of origin, FRBs can probe extragalactic medium. Radio emissions disperse through the ionised plasma between galaxies (Petroff et al. 2019), but propagation scales inversely with the square of the observing frequency. The dispersion measure (DM) is the integrated free electron column along the line of sight¹. As FRBs are detectable at cosmic scales (Macquart et al. 2020), the column is dominated by the extragalactic medium and therefore, can probe the hot, diffuse baryons (Macquart et al. 2020).

The type of morphological anomaly present in a given FRB, is not known in advance. Consequently, the data, rather than a hypothesis, was used to determine what constitutes an abnormal burst. An unsupervised search for morphological anomalies ran across 4109 catalogued sub-bursts. By using six non-parametric detectors, chosen to span the three anomaly types defined by Chandola et al. (2009), together with the subspace case (Aggarwal & Yu 2001), this work aims to understand if the morphology of any given burst, within the second release, is atypical and whether that can be established from the tabulated parameters alone.

Unsupervised and semi-supervised morphology on the second catalogue now exists. Sun et al. (2026) and Fan et al. (2026) focus on clustering within a population rather than a search for outliers; neither reports the geometries the respective methods would have missed. Injection-based characterisation is an established practice for this instrument. Merryfield et al. (2023) inject approximately 8.5×10^4 synthetic bursts to measure the detection capabilities of CHIME. The measurement made here sits downstream of that and asks what an unsupervised detector recovers once a burst is already in the catalogue.

This investigation returns a shortlist of 36 candidate bursts. A ranking detector will return a most anomalous burst (Chandola et al. 2009). Therefore, a sensitivity floor was measured for each method and anomaly type, including one geometry that no method recovered at any strength tested. Specificity testing rejected the most sensitive method due to poor discrimination of noise and signal; the resulting detector uses local outlier factor (LOF), k nearest neighbour (kNN) distance and cluster-based local outlier factor (CBLOF).

2 Method

This section will outline the morphological analysis of Catalog 2 from Collaboration et al. (2026). The data was accessed via the CANFAR data archive (The CHIME/FRB Collaboration 2026). The release holds 5045 rows covering 4539 distinct bursts. The pipeline consists of five stages: cleaning, detection, two validation passes and context analysis.

¹DM = $\int n_e dl$, with n_e the free electron density and dl a path element.

2.1 Cleaning

The first artefacts removed were the CHIME quality flags: `excluded_flag`, used for known general issues with the measurement such as instrument testing; `sidelobe_flag`, used for detections collected from sidelobes²; and `intrachan_flag`, alerting to the large instrumental smearing of an FRB. This removed 494 rows, a further 442 were removed due to missing features resulting in a total of 4109 distinct sub-bursts. The cleaned data set was labelled using `repeater_name` designating 1047 repeater sub-bursts across 81 repeater sources and 3062 non-repeaters.

Eight morphological features from each sub-burst were used to create parameter space. These were DM (`dm_fitb`), width (`width_fitb`), peak flux (`flux`) and fluence (`fluence`) (both lower limits), spectral index (`sp_idx`), spectral running (`sp_run`), peak frequency (`peak_freq`), and bandwidth. DM is a line-of-sight quantity rather than a property of the burst but was retained because it carries the environment of the source. DM drives no candidate in the shortlist. Despite being a morphological parameter, `scat_time` is recorded as zero for 58.8% of the catalogue.

The data was normalised to $\mu = 0$ and $\sigma = 1$. `dm_fitb` and `width_fitb` were log transformed as they are strictly positive and non-zero. Whereas a $\log(1+x)$ transform of `flux` and `fluence` was taken as these values can approach or equal zero. Finally, $\text{sign}(x) \log(1+|x|)$ was taken of `sp_idx` and `sp_run` as they can take positive and negative values.

Four of the eight features are not mutually independent. Peak frequency is recoverable to within 0.01% for 3464 of the 3947 retained narrow-band fitted sub-bursts from spectral index and spectral running. Bandwidth is partly determined by the same pair. The feature set was retained as published rather than reduced.

2.2 Detectors

The cleaned data was run through a set of six non-parametric detector methods and a per-sub-burst score was recorded. The six methods were chosen to cover the standard anomaly geometries of point, contextual and collective (Chandola et al. 2009) and subspace (Aggarwal & Yu 2001). Note, the anomaly scoring was blind to the `repeater_name` parameter.

LOF (Breunig et al. 2000) explores the local density principle and compares the local density of each burst to its k nearest neighbours. A value near unity indicates a burst whose local density matches that of its neighbours, whether that neighbourhood is dense or sparse. Values substantially above unity indicates a burst whose local density is low relative to its own neighbours’.

The values of k were swept over `K_VALUES` = {30, 50, 75, 100, 150}, encompassing $\sqrt{N} = 64$, which is a heuristic rule proposed for density estimation in six dimensions or fewer (Lall & Sharma 1996). The asymptotically optimal neighbour order scales as $N^{4/(d+4)}$, so in eight dimensions it anchors the sweep rather than setting it. By sweeping values of k , the ranking was found to be independent of neighbourhood size.

Extended isolation forest (EIF) (Hariri et al. 2019) addresses the isolation principle and uses 100 binary trees, and a 256-burst subsample per tree (Liu et al. 2008) to make random hyperplane cuts through eight-dimensional feature space. The fewer divisions of parameter space needed to isolate a burst suggests it may be more atypical.

kNN distance (Ramaswamy et al. 2000, Angiulli & Pizzuti 2002) probes the point and local principle, measuring the mean distance to the k nearest neighbours. The distance is absolute rather than normalised against neighbouring density allowing the method to respond to global

²The secondary weaker maxima off axis from the telescope. A sidelobe burst has a poorly constrained position and flux value.

isolation. Taking the mean of the k distances gives the same anomaly ranking as the summed distance in Angiulli & Pizzuti (2002) as the two aggregations differ by the constant factor k . The mean ordering differs from the k -th-distance formulation of Ramaswamy et al. (2000) in using all k distances rather than the largest alone. As with LOF, values of k were swept over `K_VALUES`.

CBLOF (He et al. 2003) examines the collective principle. k -means clustering (MacQueen 1967), using the k -means++ initialisation (Arthur et al. 2007), partitions all the data into twenty clusters. The clusters are sorted and divided into ‘large’ or ‘small’ by either of two rules. The first, the clusters accumulated already contains $\alpha = 0.9$ (90%) of all points; the second, the current cluster is $\beta = 5$ times larger than the next largest. Each burst was scored by the distance to the nearest ‘large’ centroid. Small clusters are far from large ones hence; this method was implemented to identify micro-clustering. This implementation omits the cluster-size weighting of He et al. (2003) and scores by distance to the nearest large-group centroid alone.

A random-subspace ensemble (RSE) (Lazarevic & Kumar 2005) was used to address the subspace and correlation-breaking principle. This method explores the data at a subset size between $d/2$ and $d - 1$, where d is the number of parameters, which for eight features is four to seven (Lazarevic & Kumar 2005). Subsets are scored independently by a base LOF detector, and the per-subset results are combined such that an anomaly confined to a feature combination not present in all dimensions is recoverable. The ensemble was drawn fifty times at the detection stage but twenty times in the injection and validation stages, reducing run time. The per-subspace ranks were mean and maximum aggregated, the two standard combination functions for outlier ensembles (Aggarwal & Sathe 2015). Ranks were used rather than raw scores because RSE scores are not comparable across differing dimensionality. Explicitly, the maximum aggregation, `subspace_max`, is a second view of this one method, not a seventh method.

Finally, a non-parametric conditional anomaly detection (CAD) system, based on Song et al. (2007), was used to investigate the context principle. Using DM as a distance proxy and peak frequency as band position, CAD creates context space and finds the k closest neighbours. Once again values of k were swept over `K_VALUES`. The anomaly score is the mean distance of a burst from those same neighbours in morphology space.

2.3 Injection Recovery

Testing the capabilities of the detection methods required injection of synthetic anomalies, of known types, into the real feature set. Sensitivity was measured as the fraction of injected anomalies ranked in the top 5% of the augmented dataset. The six anomaly types tested were global and local outliers, collective and collective resolvable micro-clustering, subspace correlation-breaking and contextual. Each type uses a parameter `strength`, which was set to take values of 0.25, 0.5, 0.75, 1.0, 1.5 and 2.0. The number of injected points per trial, `N_INJECT`, was fixed at 20 for every geometry except the resolvable collective cluster. `strength` was defined separately for each geometry.

Global anomalies scale with `strength` by the Euclidean distance from the sample mean; the local and two collective anomaly types use the mean distance to the twenty nearest neighbours. For subspace correlation-breaking, `strength` interpolates the fitted `sp_idx` against `sp_run` relationship. Contextual anomalies use the morphology distance from context-peers. At `strength` 1.0, the synthetic anomalies were scaled to match the most extreme real burst geometry. Note, as the definition of `strength` varies over anomaly type, the sensitivity floor is only comparable between methods for a given type.

Global outliers were `N_INJECT` independent points, each placed at a distance `strength` $\times$ `D_MAX` from the mean in a random direction, where `D_MAX` was the Euclidean distance from

data mean to the most extreme real burst. An additional Gaussian displacement of $\sigma = 0.05$ was added.

Local outliers were generated by taking real bursts (anchors) and displacing them by a factor of four local neighbour spacings, which was scaled by **strength**, to a region of low density. **k_local** was independent of the k sweep and each artificial anomaly was isolated and moved in an independent direction.

Collective micro-clusters were $N = 20$ injected bursts within $\sigma = 0.05$ of each other centred on an anchor, displaced by a **strength** local-neighbour spacings into a sparser region of feature space. A resolvable collective micro-cluster had a population of $N = 80$, chosen to exceed the LOF neighbourhood size used in the injection runs, $k = 75$. The resolvable collective cluster was displaced from an anchor by three local-neighbour spacings $\times$ **strength** into a sparser region but not the global tail.

Subspace correlation-breaking took N real bursts and interpolated the correlation away from the **sp_idx** and **sp_run** real values. At strength 0, pairs have normal correlation, at 0.5 the pairs were decorrelated and 1.0 had an inverted correlation. Both marginals remain within a normally distributed range. **sp_idx** was clipped to ± 1.5 and **sp_run** to ± 2 , making the break visible only in the joint distribution. At **strength** 1.5 and above, the inversion begins to push the **sp_run** marginal into an atypical tail which becomes detectable by distance.

Contextual anomalies took the context, the DM and **peak_freq**, of an anchor and linearly interpolated the remaining morphological features with that of the most context-distant (donor) of thirty candidates drawn at random from the retained sample. **strength** was the interpolation fraction from the morphology of the anchor toward that of the donor; a value of 1.0 described the synthetic burst having the morphology of the donor but the context of the anchor.

To establish a result before the sweep occurs, one augmented dataset per type was generated at a fixed **strength** of 1.0 and projected, together with the real bursts, into a two-dimensional principal component analysis (PCA) space, which was fitted on the real data.

Scoring was repeated over the six types and strengths with five repetitions, producing 180 trials. The results are collated into a table describing the mean top 5% recovery per type, strength and method, averaged over **N_REPEATS** with rows denoting type and strength and columns, the method. To bound runtime over the trials, the injection and validation stages used one fixed neighbourhood size per method, $k = 75$ for LOF, $k = 20$ for kNN and $k = 50$ for CAD, and the RSE was drawn twenty times rather than fifty.

The sensitivity floor for a given method type pair was defined as the lowest swept strength at which the mean top 5% recovery reached 0.50 (50%). If this was never reached, the floor was recorded as >2.0 . A recovery curve for each method was plotted with error bands of $\pm 1\sigma$.

2.4 Candidate Validation

Score concentration was found by sorting the anomaly scores per sub-burst into descending order and min-max normalising (Patro & Sahu 2015) per method, making the scores comparable across methods. The tail prominence, the ratio of the 20th-ranked normalised score to the median normalised score, was taken with a value near unity implying the highest-scoring sub-bursts did not form a separated tail. This was plotted, where a sharp early knee corresponds to a high tail prominence and a near-linear profile to a low tail prominence.

To understand the stability of the methods that use random feature subsets, **subspace_max**, **subspace(mean)** and LOF, the mean pairwise Jaccard overlap (Ben-Hur et al. 2001) of the top twenty anomalies was found. This was repeated over eight random seeds.

A cross-method agreement was found, comparing bursts from the top 1% of each method. The shortlist was created using bursts that passed the specificity and sensitivity tests and were flagged by at least two of the three retained detectors, LOF, kNN and CBLOF.

2.5 Anomaly Context

A single parameter cannot describe an anomaly that resides in a relationship between features. Three further diagnostic quantities were recorded alongside the shortlist: `corr_break_margin`, dominance ratio and CAD percentile. $\text{corr_break_margin} = |r_{\text{resid}}| - \max(|z_{\text{sp_idx}}|, |z_{\text{sp_run}}|)$, where r_{resid} is the residual of the fitted `sp_idx` against `sp_run` relationship and z denotes a scaled feature value. The dominance ratio is the $|z|$ of the driver, the feature carrying the largest absolute scaled value for that sub-burst, scaled by the next largest $|z|$ in the same sub-burst, measuring the anomaly flag reliance on a single feature. The CAD percentile is the conditional anomaly score of a burst relative to the full population.

The driving feature of each candidate was checked for fit reliability. The relative fit error is the absolute value with respect to the raw error. Where the ratio reaches `INSPECT_THRESHOLD` = 1, the quoted error is as large as the measurement it is attached to, and the candidate carries fit-reliability flag 1. Where the driving feature has no error column in the catalogue, it carries flag 2. Otherwise, it carries flag 0. No candidate was dropped based on this annotation.

The full implementation of the pipeline, including data retrieval, preprocessing, the six detectors and injection generators and verification as well as the resulting visualisation is available on GitHub at `FRBAnomalyDetection` (Cox 2026).

3 Results

This section outlines the results, detailing the performance of each detection method and the results of the sensitivity and specificity testing.

Figure 1 describes the projection of each of the injected anomaly types compared to the axes of greatest variance in the real feature space, PCA 1 and 2. Global and local anomalies sit outside the bulk, the other types are not separable in two-dimensional feature space. Recovery was measured as a function of injected strength. Figure 2 illustrates the top 5% recovery for each method against each geometry. The sensitivity floor was reported at the value of `strength` at which the recovery curve crosses 0.50.

Of the retained detectors used for the shortlist, LOF and kNN had the highest sensitivity scores, both reaching 0.50 on global anomalies, 0.50 and 0.75 respectively on local anomalies, and both had a floor of 1.50 on subspace correlation-breaking and contextual anomalies. Neither scored below 2.0 on the two collective geometries. No method recovered the collective micro-cluster. The resolvable micro-cluster was recoverable by CBLOF at strength 1.00. EIF and CAD were excluded on sensitivity, having been outperformed on every geometry by a retained detector (see Appendix B). The two RSE variants cleared the sensitivity stage but were rejected at the specificity stage. The set of detection algorithms able to recover the majority of injected anomalies are LOF, kNN and CBLOF, hence, the shortlist was found using these detectors. The full set of sensitivity scores for all seven variants can be found in Appendix B.

The internal spread of real repeating sources, within feature space, was measured directly. Across the 35 repeating sources contributing at least five sub-bursts (938 sub-bursts in total), the median per-source standard deviation is 0.50 in the eight scaled features and 0.57 with DM excluded³. The median is stable against the choice of threshold, moving between 0.48 and 0.60 as the minimum source size is varied from two sub-bursts to twenty. The tightest source measured is 0.23. Figure 4 shows whether the highest-scoring bursts form a separated group or sit at the top of a smoothly graded ranking. The mean pairwise Jaccard overlap under eight seeds for `subspace_max` was 0.52 and 0.63 for the mean aggregation. LOF is deterministic, thus,

³A repeating source returns a nearly constant DM. Its median within-source standard deviation is 0.003 scaled units, which lowers the eight-feature mean without describing any morphological similarity between the bursts.

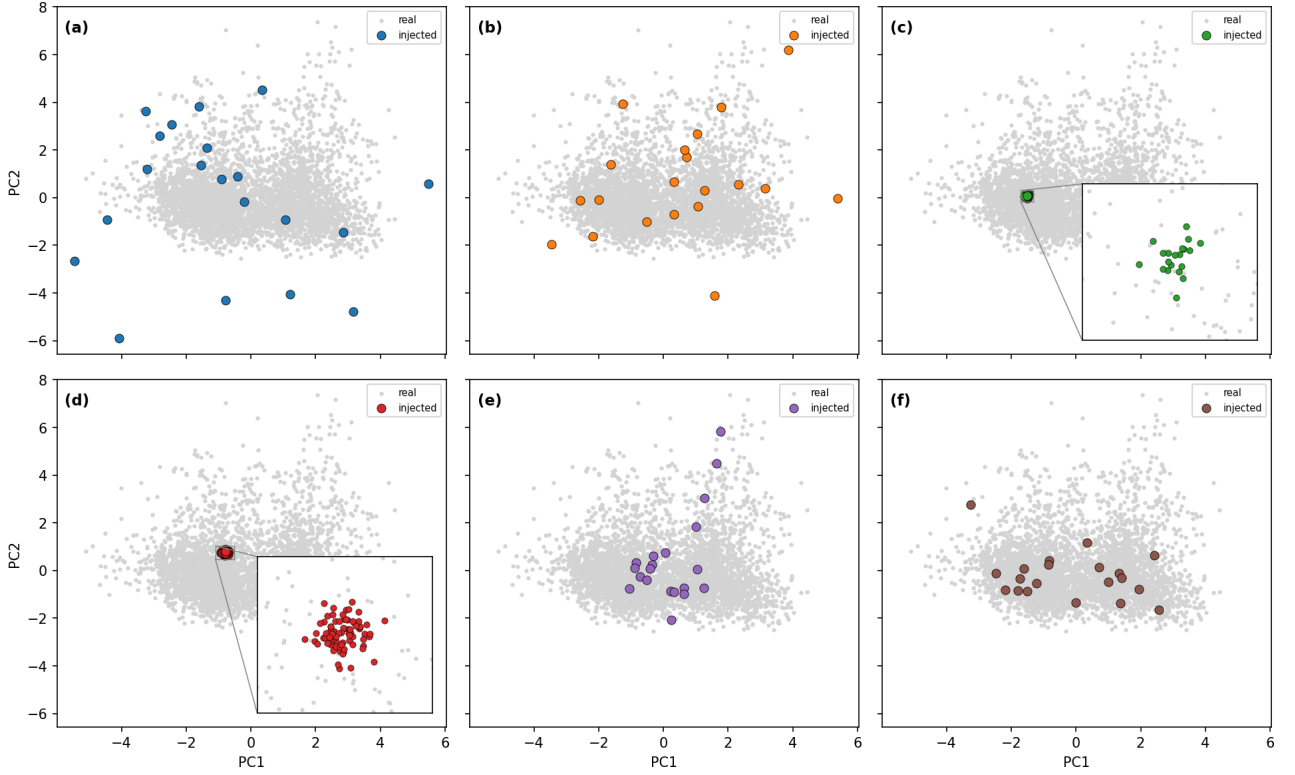


Figure 1: Two-dimensional PCA space with each injected anomaly class, visualised using the greatest variance in the real feature space. Panels (a) to (f) are global, local, collective, resolvable collective, subspace correlation-breaking and contextual anomalies respectively.

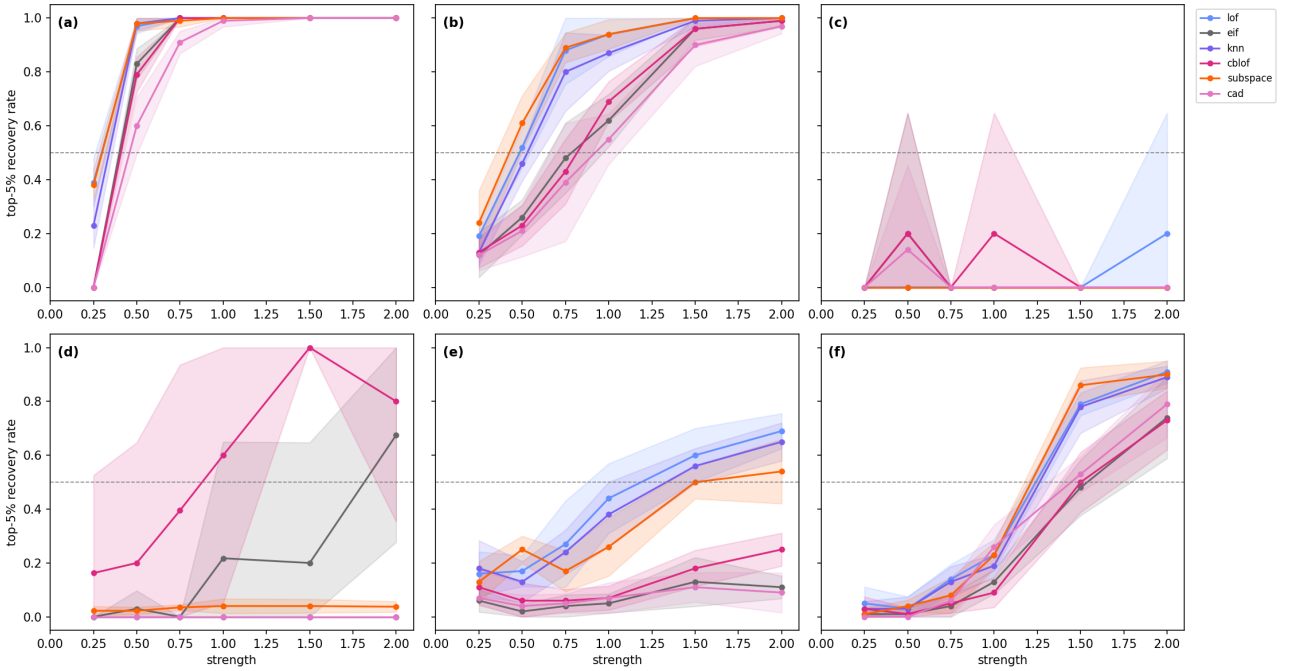


Figure 2: Injection-recovery rate of the top 5% of anomalies against anomaly strength, with errors to $\pm 1\sigma$. Panels (a) to (f) are the performance of each method on global, local, collective micro-clustering, resolvable collective micro-clustering, subspace correlation breaking and contextual anomalies, respectively.

has a value of 1.00. Note, neither criterion has an externally defined limit. Therefore, selection was relative across the seven variants and `subspace_max` ranks last on both specificity tests.

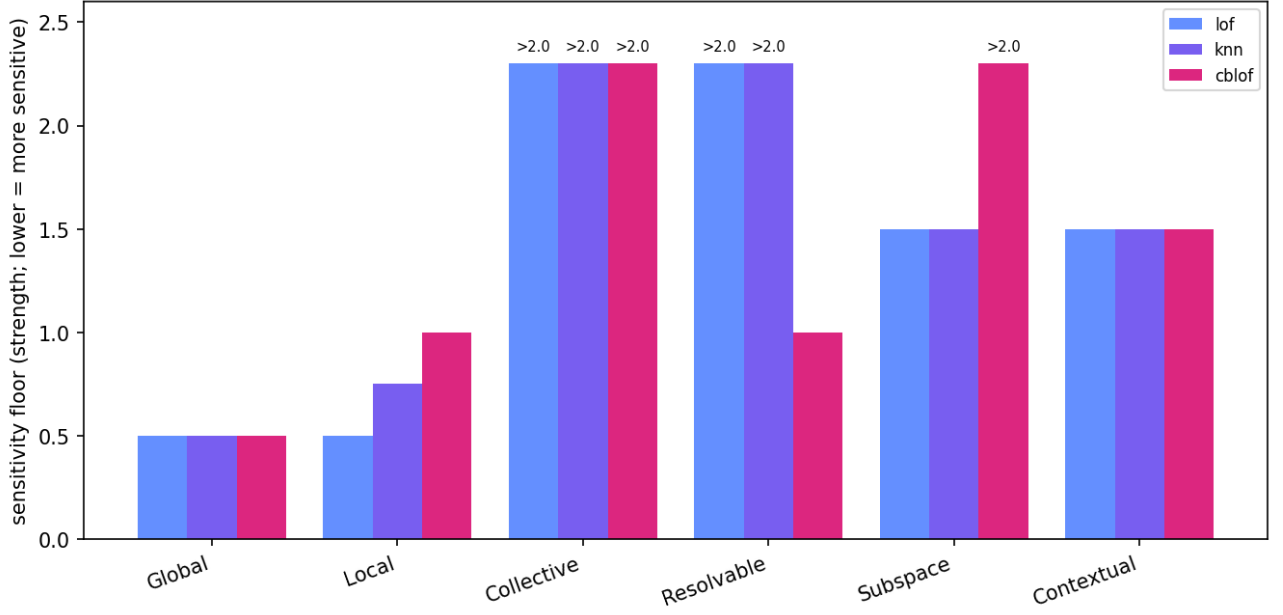


Figure 3: Sensitivity floor of the three final detection algorithms, LOF, kNN and CBLOF. >2.0 implies the mean top 5% recovery never reached 0.50 at any strength in the sweep. Floors are comparable between methods within the same anomaly type.

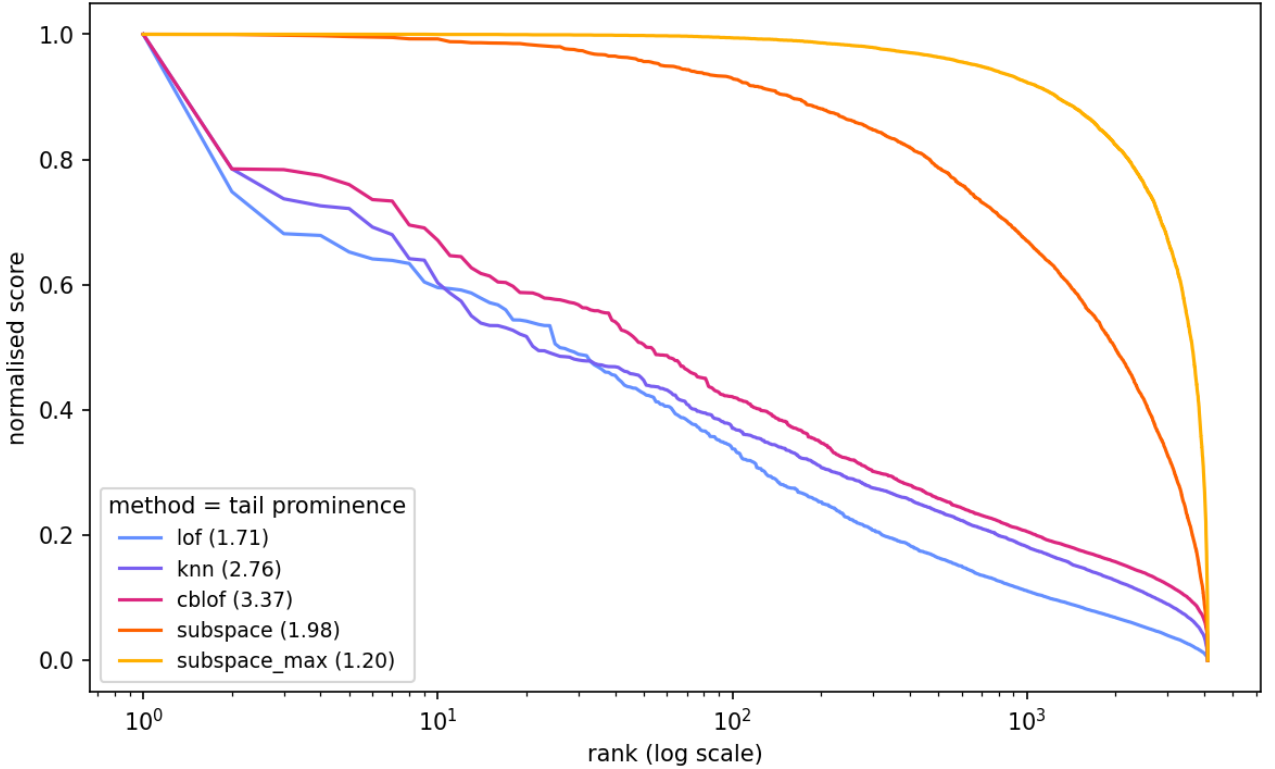


Figure 4: Normalised score concentration of LOF, kNN and CBLOF. Averaged and maximal RSE aggregation used for comparison. The tail prominence was also included.

The shortlist contains 36 flagged sub-bursts from 35 sources, of which, 29 sources are non-repeaters, and six are from repeating sources, FRB20220912A produced two of the listed sub-bursts (FRB20230826A and FRB20221219D).

TNS Name	Sub	N	Driver	Err	Max Rel Err	Dom. Ratio	Corr. Margin	CAD %	Flag
FRB20220407A	0	3	width_fitb	38.41	38.41	7.42	-0.23	1.00	1
FRB20210423B	0	3	width_fitb	8.16	8.16	2.76	-0.30	0.99	1
FRB20211108A	1	3	width_fitb	5.74	5.74	5.25	-0.05	1.00	1
FRB20200127A	0	3	width_fitb	3.66	3.66	2.08	-0.10	0.99	1
FRB20200130B	0	3	width_fitb	2.17	2.17	2.20	0.22	0.98	1
FRB20201126C	0	3	flux	0.60	0.60	1.73	-0.45	1.00	0
FRB20230826A	1	3	width_fitb	0.57	0.57	2.83	-0.28	0.99	0
FRB20200429A	0	3	sp_run	0.47	0.61	1.10	-0.35	0.99	0
FRB20210512C	0	3	sp_run	0.39	0.61	1.08	-0.36	0.99	0
FRB20200705D	0	3	sp_run	0.38	2.81	1.17	0.61	0.98	0
FRB20210222C	0	3	sp_run	0.35	0.93	1.10	-0.44	0.99	0
FRB20190104B	0	3	sp_run	0.28	0.76	1.10	-0.26	0.97	0
FRB20200723B	0	3	fluence	0.26	0.29	3.01	-0.84	1.00	0
FRB20221219D	0	3	sp_run	0.11	0.63	1.38	-0.90	0.99	0
FRB20200122B	1	3	width_fitb	0.03	7.42	1.66	1.74	0.98	0
FRB20210513C	0	3	peak_freq		2.33	1.13	-0.13	0.99	2

Table 1: Reduced shortlist where all three methods flagged the burst as anomalous (see Appendix A for the full shortlist). As presented, the data is ordered by agreement count and then by driver relative fit error. **TNS name** is the name of the burst, **Sub** is the sub-burst index within a burst that was fitted with more than one component. **N** describes how many methods independently found a burst to be an outlier. **Driver** denotes the feature with the largest $|z|$ score and **Err** describes the relative error for the measurement. **Max Rel Err** is the largest error on any feature within the flagged candidate. **Dom. Ratio** is the $|z|$ score of the **Driver** feature relative to the next highest $|z|$ score. **Corr. Margin** tests for subspace correlation breaking, positive implies this type of anomaly is the cause of the flag. **CAD%** is the CAD score of the burst relative to the full population; higher implies contextually the burst is more anomalous. **Flag** is the fit-reliability verdict. Flag 0, the relative fit error of the driver is below unity; flag 1, it is at or above unity, so the stated error is at least as large as the value it is attached to; flag 2, the driver has no error column in the catalogue.

Repeaters occupy a dense region of the feature space. Across the retained sub-bursts, 69.9% of the 20 nearest neighbours to a repeater are themselves repeaters, against a catalogue baseline of 25.5%. That figure is inflated by within-source clustering, since 47.1% of repeater neighbours are other sub-bursts from the same source. Excluding them leaves 36.9%, still above the baseline. For non-repeaters, the mean repeater fraction is 10.6%. One non-repeater in the shortlist, FRB20181119D, resides within a region densely populated by repeaters. The second sub-burst returns 15 repeater neighbours from 8 distinct repeating sources. Note, the repeating fraction of the shortlist is 19.4%, which is not distinguishable from the catalogue baseline. Across the 4109 retained sub-bursts, 26.2% have spectral running, flux or fluence as the largest absolute scaled value. Within the shortlist this population is 52.8%. A one-sided hypergeometric test against the population (Fisher 1935) gives $p = 5.8 \times 10^{-4}$ with an odds ratio of 3.18, equivalent to 3.25σ .

By fit reliability within the shortlist, 25 sub-bursts have a flag 0, 9 a flag 1 and 2 carry flag 2. The drivers are dominated by width (13 candidates) and spectral running (9), followed by flux (6), fluence (4), peak frequency (2) and spectral index (2). FRB20220407A ranks first on all three methods and carries flag 1, with a fitted width error 38 times the measurement itself. **width_fitb** is the intrinsic burst width returned by the catalogue fit, deconvolved from instrumental smearing and scattering, rather than an observed duration. Note, a width close to the instrumental time resolution carries a large fractional error by construction (Collaboration et al. 2026). The flag records an unconstrained fit rather than a rejection candidate.

Bandwidth, DM and fitted width reach maximum absolute scaled values of 1.98, 3.07 and 8.08 respectively. Across the retained population of the catalogue, bandwidth is the largest

absolute scaled value of 735 sub-bursts and DM, 681.

4 Discussion

Six non-parametric detectors, validated by anomaly injection, return a shortlist of 36 atypical candidates, each annotated with the parameter driving the flag. By finding a morphologically atypical burst, the population itself can be better characterised, with outliers providing the boundary for normality.

FRB morphology has often been used to characterise a population rather than to isolate its members. Pleunis et al. (2021) treat burst shape as evidence to understand whether repeating and non-repeating sources share an origin. The unsupervised work on the second release clusters the sample rather than searching it (Sun et al. 2026, Fan et al. 2026). But Kuiper et al. (2025) report a continuum rather than discrete morphological classes in the first CHIME release and name training-set size as a limitation. This work sought to understand whether the morphology of any single burst is itself atypical, not whether the population divides into groups or whether repeating and non-repeating sources share an origin and returns a shortlist rather than a partition.

Sensitivity is not a single property of a detector but conditional on anomaly type. The synthetic anomalies were designed to span a variety of geometries, rather than one generic outlier, so that a blind spot could be attributed to a shape rather than to a strength (Chandola et al. 2009, Aggarwal & Yu 2001). The global anomalies carry an isotropic displacement of $\sigma = 0.05$. This meant the twenty bursts, injected at the same strength, did not form a degenerate shell at a single radius, which the density methods could read as a micro-cluster rather than as a set of global outliers. The local generator displaced the synthetic anomalies by a sufficient number of local neighbour spacings as to not form an in-population micro-cluster. A cluster displaced into the global tail stops testing collective structure and starts testing global isolation, which every variant tested recovers (see Appendix B).

For the contextual anomalies, rather than drawing from the single globally most context-distant burst to become donor, the most context distant of thirty randomly drawn donors was chosen. This reduced the probability of biasing the contextual synthetics to one fixed extreme reused across every draw. `k_local` was held at twenty, independent of the `K_VALUES` sweep such that the injection geometry would not move with the detector configuration.

Despite `subspace_max`, the max-aggregated view of the RSE, having the lowest sensitivity floor on four of the six synthetic geometries (see Appendix B), it was rejected for the final detector suite. Both the specificity checks, mean pairwise Jaccard overlap and tail prominence score, described an indiscriminate detector. By taking the maximum aggregation over a large enough sample, the upper bound over all subspaces will be returned and therefore, the statistic tends to overestimate (Zimek et al. 2014). The mean aggregated RSE had a more prominent tail septation than the retained LOF but failed stability testing and was therefore, rejected too. Sensitivity and specificity establishes that the retained detectors produce rankings driven by structure in the data rather than noise. But it does not confirm that any individual flag is astrophysically real. It was established in this work that the tabulated parameters could not make that clarification.

The two specificity criteria were not fixed before the real-data scores were inspected. Both are standard measures in the outlier-ensemble literature (Ben-Hur et al. 2001, Zimek et al. 2014), applied identically across all seven variants rather than as thresholds tuned to remove one method in particular. The ordering reflects the data rather than a criterion built around a known result.

An exhaustive understanding of outlier types cannot be known in advance (Steinbuss &

Böhm 2021). Therefore, the six detectors were chosen to have differing inductive biases, to better respond to structure that was never injected. A further three methods were explored but ultimately rejected due to the shape of morphology parameter space. A Gaussian mixture and its conditional variant (Dempster et al. 1977) were tested and dropped before injection as both require a describable reference distribution with a fixed number of elliptical cores. No such core exists within the CHIME catalogue; the features are multimodal with heavy tails. Furthermore, fitting either requires choosing several components, and the Bayesian information criterion (Schwarz 1978), which trades goodness of fit against parameter count and whose minimum conventionally sets that number, has no minimum for any covariance type between one and twenty components for this parameter space. The heavy tails absorb each further component rather than converging the score. Robust Mahalanobis distance under a minimum covariance determinant estimate uses the same elliptical assumption (Rousseeuw & Driessen 1999). It is correlation aware, unlike the Gaussian mixture, but that capability turned out not to be distinctive. Both LOF and kNN are able to detect the correlation-breaking geometry (see Appendix B) without fitting an ellipsoid to a distribution that is not elliptical.

Spectral running, flux and fluence drive the majority of flagged bursts. However, the three are not independent routes into the tail. A burst detected through a grating lobe, an off-axis maximum of the formed beam, has part of the spectrum truncated. So, a broadband burst is recorded as narrowband, and Collaboration et al. (2026) attribute the bimodality in spectral running among non-repeaters primarily to that effect. Further, a burst detected away from the beam centre is recorded as fainter, which is why the catalogue publishes flux and fluence as lower limits. The shortlist enrichment is likely due to the same systematic error counted three times.

Large absolute values of spectral running are characteristic of the narrowband spectral shape the running power law was written to describe (Pleunis et al. 2021, Fonseca et al. 2024). The parameter is also model-dependent; Collaboration et al. (2026) report that the `fitburst` model was revised between the first and second releases following a change to the `fitburst` model thus, an extreme value can reflect the fit as much as the burst. The data cleaning applied to this work cannot separate the model fit changing from an indirect detection. `sidelobe_flag` removes only 37 far-sidelobe events, identified by “spiky” spectra (Collaboration et al. 2026), so near-sidelobe detections bypass the flag. Comparing the driver distribution of the flagged rows against the retained population could resolve this. However, all bursts with `sidelobe_flag` have at least one morphological feature missing, so cannot be compared by the models. Therefore, the distinction between a narrowband burst and a burst with a poorly constrained spectral fit can only be separated by the dynamic spectrum. Note, the beam dependence itself is documented in the catalogue release rather than found here.

The degeneracy between intrinsic width and scattering cannot be broken from the tabulated columns, since scattering time is absent from the feature set. Collaboration et al. (2026) omit the scattering parameter where an F-test at the 0.1% level does not favour a free-scattering model over a fixed-zero one. Thus, a zero encodes a non-detection rather than a measured absence of scattering. The width-driven candidates are unlikely to be astrophysically anomalous because a width that approaches the instrumental time resolution retains little signal after deconvolution. The width driven bursts were retained as they describe where the sensitivity of the detector is set by resolution rather than by morphology.

Cluster size is part of what defines a collective anomaly, so `N_INJECT` was fixed for every geometry except the resolvable collective. By retaining the weighting of He et al. (2003), the score of a burst inside a large, injected cluster is suppressed proportional to the size of said cluster, which is contradictory to the behaviour the injection was designed to measure. No method recovered the smaller cluster at any injected strength, and only CBLOF meaningfully recovered the larger one.

At $k = 75$, used in the injection runs, the resolvable cluster was larger than the LOF neighbourhood. LOF never reached past the synthetic points into real data so scored the synthetic cluster as ordinary density. CBLOF recovers the same cluster because k -means partitions it as a cluster, which puts its members far from the nearest large centroid. At $k = 20$ the kNN neighbourhood and the smaller cluster are the same size, so every member has its nineteen fellow members as its own neighbours and returns a small mean distance. Twenty points also fall below the size at which k -means separates the group into a cluster of its own. Hence, detection of a collective anomaly requires a cluster large enough for a clustering method to resolve it, and small enough that a density method reaches beyond it into real data.

The smaller cluster was injected with a scatter of $\sigma = 0.05$, an order of magnitude tighter than the median within-source spread measured across real repeating sources. That value was fixed before the within-source spread had been measured and sits well within anything the catalogue contains. A synthetic group this tight is more internally distinct than any real source measured, so its failure to be recovered cannot be attributed to the knot being insufficiently separated from the bulk. Rescaling the injection to the real within-source spread would not test the detectors further, because a twenty-point group at that scatter carries a density contrast of order unity against the surroundings and is no longer a local over-density. The blind spot demonstrated here is a property of the feature space rather than of the detectors.

CAD was not retained in the final detector, despite being built specifically to find contextual anomalies. Context space was constructed from DM and peak frequency. However, peak frequency is recoverable from spectral index and spectral running, both of which sit in the behaviour space. The catalogue does carry candidate context parameters outside the feature set, including sky position, exposure and signal to noise. But the injection system defines contextual anomalies within the same space the detectors score, so widening context space would have required widening the feature set itself. Of the eight tabulated parameters, DM is the only one that is influenced by environmental features rather than a description of the shape of the burst alone, so the catalogue does not offer a second axis for CAD to use that is not itself a behaviour feature. This is why each of the 36 candidates carries a high CAD percentile (see Appendix A). Whereas EIF was excluded as it did not outperform any of the retained detectors. EIF is designed to identify global point anomalies, separated from the bulk, but failed to outperform other, less specialised detectors.

Any burst whose time or frequency dimensions were anomalous will have been missed as these properties are not present within the tabulated parameters. Similarly, bandwidth and DM, whilst present in this work, drive no candidate in the shortlist. Both parameters reach maximum absolute scaled values below width, so neither reaches a magnitude that places a sub-burst in the tail of the distribution. Therefore, the shortlist could be read as a search over the six features that can carry an extreme scaled value.

If the shortlist was reduced by removing bursts with drivers which were beam-affected, carried a larger fit error than the measurement or driven by the beam dependent spectral index or peak frequency, four would remain. This does not detract from the work as it is a shortlist noted for further inspection rather than a finding of anomalous bursts. The tabulated parameters cannot confirm an astrophysical origin nor a systematic error. Removing the potentially instrumentally driven bursts would assert a systematic explanation on the same evidence that is insufficient to assert an astrophysical anomaly. Where previous searches for morphological atypicalness have partitioned the population rather than searched it for individual outliers, this work identifies 36 potentially anomalous FRBs, of which four survive filtering on the known instrumental effects.

FRB20181119D is an apparent non-repeater whose second sub-burst sits within a highly repeater populated region of parameter space. Whilst this work has not set out to find potential repeaters, the presence of FRB20181119D is interesting as it could suggest burst morphology

carries information about repetition. Sub-burst FRB20181119D is flagged on spectral index, with spectral running and fitted width at comparable scaled values. The two spectral values are not independent evidence, since the pair is strongly correlated across the retained sample. The fitted width of this sub-burst shares the characteristics a typical burst that has been scattered would produce. The tabulated dataset cannot discern the true reason, making this sub-burst an ideal candidate dynamic spectrum analysis.

Originally, repeater recovery provided the validation of the pipeline. Within the first catalogue release repeaters form a small minority (Collaboration et al. 2021) and are morphologically different to the more populous non-repeater. However, within the second release, repeaters occupy a dense region of the feature space and consequently cluster membership would have been measured rather than detector sensitivity.

This work has found that a detector selected on injection recovery alone can be substantially noise-driven on real data. This can be mitigated through specificity testing. Catalog 2 is the largest FRB sample released to date, and a shortlist of individually atypical bursts built from its tabulated parameters has not been measured before. However, it has also been shown that the tabulated parameters cannot decipher beam-position enrichment and the width-resolution. A semi-supervised convolutional autoencoder on Catalog 2 dynamic spectra already exists (Fan et al. 2026), so this work adds a target list for a morphological anomaly search of FRBs.

5 Conclusion

Introduced here is a search for morphological anomalies in the tabulated data of the second CHIME/FRB catalogue release, applying six non-parametric detectors with different inductive biases. Injection recovery was used to establish the sensitivity and discriminatory abilities of each detector on synthetic anomalies. By analysing 4109 catalogued sub-bursts, the primary results of this pipeline are as follows.

- Returned a 36-candidate annotated shortlist of morphologically anomalous FRBs.
- Tabulated parameters cannot establish whether a flagged burst is astrophysically atypical or an artefact of beam position and fit resolution.
- Filtering the shortlist on beam-affected drivers and on fits whose error exceeds the measurement leaves four candidates.
- Validated an unsupervised search of the second CHIME/FRB release tabulated catalogue by injection of six anomaly types.
- Shortlist enriched by a factor of 2.0 in the three parameters dependent on beam position, 52.8% against 26.2% of the population at 3.25σ .
- Score concentration and seed stability rejected `subspace_max`, the most sensitive of the seven variants tested, on a mean pairwise Jaccard overlap of 0.52 across eight seeds and the lowest tail prominence of any variant.
- Collective anomalies must be simultaneously large enough for a clustering method to resolve it as its own cluster and small enough to not saturate a density method's neighbourhood.
- No method tested recovered a twenty-burst micro-cluster, with internal scatter 0.05σ , at any injected strength.

- Median within-source spread of the scaled repeating sources are an order of magnitude larger than the injected collective micro-cluster.
- Shortlist used LOF, kNN and CBLOF algorithms.
- Context space is not independent of behaviour space within this morphology.
- Found 7 of 36 candidates to be known repeaters, consistent with the population rate.

Beyond the shortlist, this work demonstrates morphological anomaly detection may be feasible since the release of the latest CHIME catalogue. Where earlier work has classified FRB populations or learned a representation of it, this work searched for individually atypical members (Pleunis et al. 2021, Kuiper et al. 2025). The tabulated parameters do not carry enough information for an astrophysical or systematic conclusion on the shortlist to be made. The candidates are therefore, offered as a target list for a morphological search of the dynamic spectra.

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Appendix A

Table A.1 lists all 36 candidates, ordered by agreement count and then by the relative fit error of the driving feature.

TNS Name	Sub	N	Driver	Err	Max Rel Err	Dom. Ratio	Corr. Margin	CAD %	Flag
FRB20220407A	0	3	width_fitb	38.41	38.41	7.42	-0.23	1.00	1
FRB20210423B	0	3	width_fitb	8.16	8.16	2.76	-0.30	0.99	1
FRB20211108A	1	3	width_fitb	5.74	5.74	5.25	-0.05	1.00	1
FRB20200127A	0	3	width_fitb	3.66	3.66	2.08	-0.10	0.99	1
FRB20200130B	0	3	width_fitb	2.17	2.17	2.20	0.22	0.98	1
FRB20201126C	0	3	flux	0.60	0.60	1.73	-0.45	1.00	0
FRB20230826A	1	3	width_fitb	0.57	0.57	2.83	-0.28	0.99	0
FRB20200429A	0	3	sp_run	0.47	0.61	1.10	-0.35	0.99	0
FRB20210512C	0	3	sp_run	0.39	0.61	1.08	-0.36	0.99	0
FRB20200705D	0	3	sp_run	0.38	2.81	1.17	0.61	0.98	0
FRB20210222C	0	3	sp_run	0.35	0.93	1.10	-0.44	0.99	0
FRB20190104B	0	3	sp_run	0.28	0.76	1.10	-0.26	0.97	0
FRB20200723B	0	3	fluence	0.26	0.29	3.01	-0.84	1.00	0
FRB20221219D	0	3	sp_run	0.11	0.63	1.38	-0.90	0.99	0
FRB20200122B	1	3	width_fitb	0.03	7.42	1.66	1.74	0.98	0
FRB20210513C	0	3	peak_freq		2.33	1.13	-0.13	0.99	2
FRB20211105D	0	2	width_fitb	8.34	8.34	2.25	-0.87	0.98	1
FRB20201120B	0	2	width_fitb	4.37	4.37	2.22	-0.03	0.98	1
FRB20190912F	0	2	width_fitb	4.09	4.09	2.52	-1.47	0.99	1
FRB20220124A	0	2	width_fitb	3.01	3.01	3.63	0.47	0.99	1
FRB20190531E	0	2	sp_run	0.90	35.93	1.00	1.37	0.96	0
FRB20220219A	0	2	width_fitb	0.70	0.70	1.69	0.45	0.97	0
FRB20190202A	0	2	flux	0.42	0.44	1.80	-0.23	1.00	0
FRB20190116C	0	2	flux	0.41	0.41	1.79	-0.20	1.00	0
FRB20190816B	0	2	flux	0.35	1.28	1.47	-0.58	1.00	0
FRB20210909B	0	2	flux	0.34	0.34	2.04	0.29	1.00	0
FRB20210415C	0	2	fluence	0.29	0.29	1.10	-0.60	1.00	0
FRB20210206A	1	2	width_fitb	0.29	0.33	1.65	0.22	0.98	0
FRB20190216A	0	2	fluence	0.28	0.69	1.11	-0.42	0.96	0
FRB20220222B	0	2	fluence	0.26	0.31	1.31	-0.37	1.00	0
FRB20190929C	1	2	sp_run	0.20	0.57	1.07	-2.21	0.97	0
FRB20181119D	1	2	sp_idx	0.20	0.61	1.04	-2.54	0.99	0
FRB20230422A	0	2	sp_run	0.20	0.79	1.37	-0.89	0.95	0
FRB20210925A	0	2	flux	0.19	0.20	1.13	-1.00	0.96	0
FRB20201119A	0	2	sp_idx	0.12	0.59	1.13	-2.84	0.92	0
FRB20180925A	0	2	peak_freq		3.78	1.02	0.51	0.98	2

Table A.1: The full 36-candidate shortlist, of which Table 1 reproduces the 16 three-method rows. Columns are as defined there.

Appendix B

Table B.1 gives the sensitivity floor for every method variant across all six injected geometries, including the four variants that did not enter the final detector.

Method	global	local	collective	resolvable	subspace	contextual
CAD	0.50	1.00	>2.0	>2.0	>2.0	1.50
CBLOF	0.50	1.00	>2.0	1.00	>2.0	1.50
EIF	0.50	1.00	>2.0	2.00	>2.0	2.00
kNN	0.50	0.75	>2.0	>2.0	1.50	1.50
LOF	0.50	0.50	>2.0	>2.0	1.50	1.50
subspace	0.50	0.50	>2.0	>2.0	1.50	1.50
subspace_max	0.25	0.25	>2.0	>2.0	0.25	1.00

Table B.1: Sensitivity floor for every method variant across the strength sweep, defined as the lowest swept strength at which mean top 5% recovery reaches 0.50 and recorded as >2.0 where no swept strength reaches it. Floors are comparable within a column and are not comparable across columns. `subspace_max` is the maximum aggregation of the RSE rather than an independent seventh method, which is why seven rows appear here against six methods in Figure 2.